

Fig. 2: An airbag deployment system

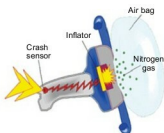


Fig. 3: An airbag deployment system after crash

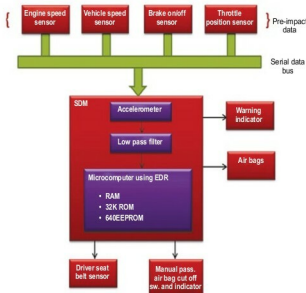


Fig. 4: Block diagram of an event data recorder

something is placed on the passenger seat, there is a change to that capacitance value. Change in capacitance value is what electrostatic capacitance sensor occupant detection system uses to determine whether the passenger seat airbag will be on or off.

Event data recorder

An event data recorder is a device installed in automobiles to record information related to vehicle crashes or accidents. It is also known as a black box. The sensing and diagnostics module, which is controlled by a microprocessor, has multiple functions, as given below:

1. Determines if a severe enough impact has occurred to warrant deployment of the airbag
2. Monitors the airbag's components

3. Permanently records information

Event data recorders record a wide range of data, including whether brakes were applied, speed and time of impact, steering angle and whether seat belt circuits were shown as buckled or unbuckled at the time of the crash.

Anti-lock braking system

Up until the 1970s, hitting the brakes too hard could lead to an accident. If the coefficient of slip between tires and the road was too low, hitting the brakes could lead to wheel lock-up. The vehicle would no longer be steerable and would start skidding. The danger is present specially when:

1. Roads are wet and slippery.
2. There are different levels of grip between the tires and the road.

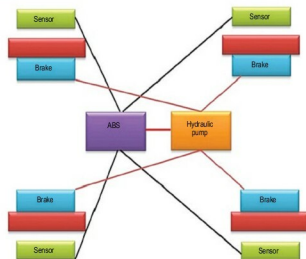


Fig. 5: A basic anti-lock brake system

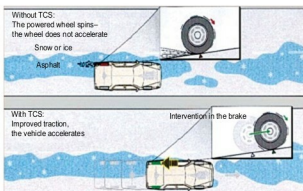


Fig. 6: Functioning of a traction control system with brakes

Wheel-speed sensors detect whether a wheel is showing a tendency to lock-up. In case it is, the electronic control unit reduces the braking pressure individually at the wheel concerned. High-speed correction of the braking pressure takes place before the lock-up threshold. The brake-fluid return together with closed-loop brake circuits makes this a safe, reliable and cost-effective system.

An anti-lock brake system is advantageous as:

1. Vehicle remains steerable even during panic braking
2. Shorter stopping distances on all road surfaces

Dynamic traction and stability control

The traction control system is required to prevent driver error from